29-04-26

Exercise 2 – SQL Aggregates and Operators

1. SELECT DISTINCT department
   FROM students;

Students

department
IT
HR
FINANCE

2. SELECT department,
   AVG (age) AS avg_age
   FROM students;

Students

department	avg_age
IT	20.5
HR	22
FINANCE	23

3. SELECT department,
   COUNT (student_id) AS student_count
   FROM Students
   GROUP BY department
   HAVING COUNT (student_id) > 1;

students

department	student_count
IT	2
HR	2

4. SELECT student_id,
name,
age,
department

FROM students

WHERE age BETWEEN 21 AND 23;

students

Student_id	name	age	department
2	Alice	22	HR
3	Bob	21	IT
4	Charlie	23	Finance
5	Eve	22	HR

5. SELECT student_id,
name,
age,
department

FROM Students

WHERE department IN ('IT', 'HR')

AND age > 21;

Students

student_id	name	age	department
2	Bob	22	HR
5	Eve	22	HR

6. SELECT department,
sum(credits) AS total_credits
FROM courses
GROUP BY department
HAVING sum(credits) > 5;

courses

department	total_credits
IT	11

7. SELECT *
FROM courses
WHERE credits != 4;

courses

course_id	course_name	department	credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

8. SELECT course_id,
course_name,
credits
FROM courses
ORDER BY credits DESC
LIMIT 3;

courses

course_id	course_name	credits
102	Python	4
103	Data Science	4
101	SQL Basics	3

9. SELECT MAX(grade) AS max_grade,
MIN(grade) AS min_grade,
AVG(grade) AS avg_grade.
FROM enrollments;

~~10. SE~~

enrollments

max_grade	min_grade	avg_grade
90	78	84,6

10. SELECT course_id,
COUNT(student_id) AS enrollment_count
FROM enrollments
GROUP BY course_id;

course_id	enrollment_count
101	1
102	1
103	1
104	1
105	1

11. SELECT department,
sum(salary) AS total_salary,
sum(bonus) AS total_bonus
FROM salaries
GROUP BY department;

Salaries

department	total_salary	total_bonus
IT	122000	10500
HR	109000	7500
Finance	70000	6000

12. SELECT department,
AVG(salary) AS avg_salary
FROM salaries
GROUP BY department
HAVING AVG(salary) > 55000;

Salaries

department	avg_salary
IT	61000
FINANCE	70000

13. SELECT employee_id,
name,
salary,
bonus,
Sum (salary + bonus) AS total_compensation
FROM salaries
WHERE Sum(salary + bonus) > 60 000;

Salaries

employee_id	name	salary	bonus	total_compensation
1	Tom	60000	5000	65000
3	Spike	70000	6000	76000
4	Tyke	62000	5500	67500

14. SELECT department,
Sum (budget) AS total_budget,
AVG (budget) AS avg_budget
FROM projects
GROUP BY departments
HAVING AVG(budget) > 70000;

Projects

department	total_budget	avg_budget
IT	270 000	135 000
Finance	80 000	80 000

15. SELECT project_id,
project_name,
department,
budget
FROM projects
WHERE budget BETWEEN 50 000 AND 120 000
NOT HAVING department ('Marketing');

Projects

Project_id	project_name	department	budget
1	AI App	IT	120000
2	Payroll system	Finance	80 000
5	HR Portal	HR	50000